



Høyskolen
Kristiania

Internet

What is the internet?

- An interconnection of billions of computers
- The Internet is available to “everyone”
- Simple user interfaces, but not quite so simple behind the scenes
- The Internet is a set of standards for network communication that interconnects different LANs and WANs, and is based on the "TCP/IP stack"
- There are 22,135 such standards, or "application layer" protocols - today we will look at 4 of them :-)

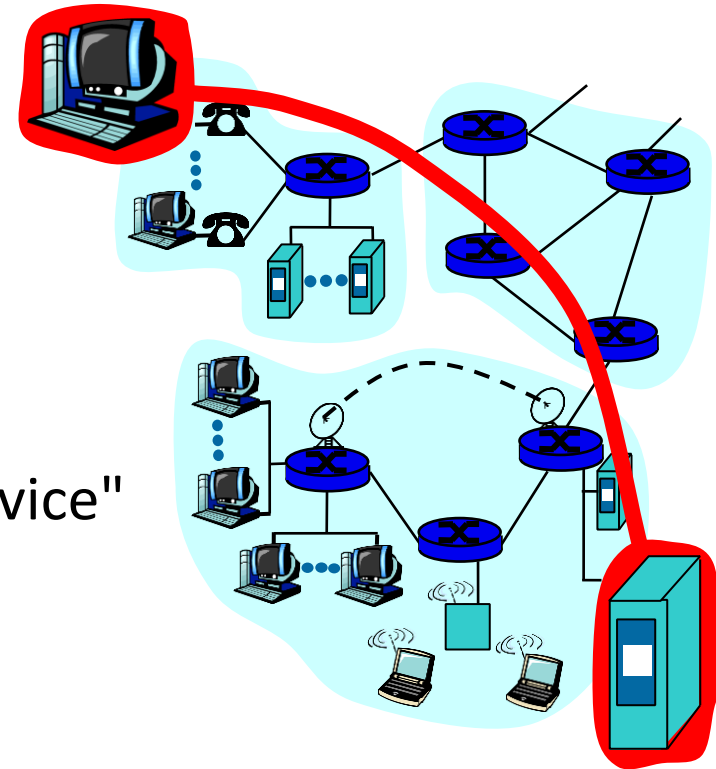
The network's outer edges

End systems (hosts)

- Running **applications**

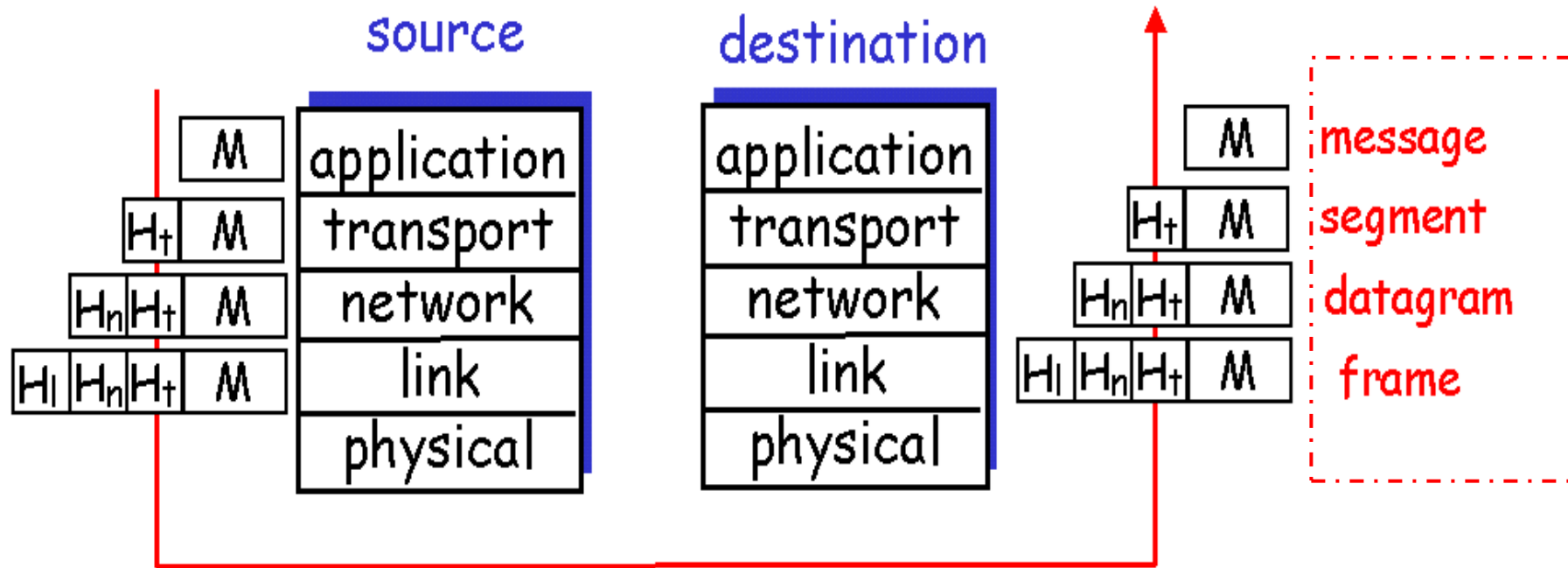
Service models:

- **Client-server** model
 - Requests come from clients that receive "service" from a server
 - For example, web browser and web server
- Peer-to-peer (P2P) model
 - All machines are equal
 - For example: BitTorrent, Skype,...



Navn på laget	Betegnelse på overføringsenhet	Viktigste oppgaver/funksjoner Eksempel på protokoller/standards
Applikasjonslaget	Melding (Message)	Støtte nettverksapplikasjoner Ex: HTTP, DNS, FTP, SMTP, POP3....
Transportlaget	Segment	Transport av applikasjonslagsmeldinger mellom klient- og tjener-sidene til en applikasjon; herunder mux/demux, ulike nivåer av pålitelighet med mer Ex: TCP, UDP,..
Nettverkslaget	Datagram	Routing av datagram fra/til vertsmaskin gjennom nettverkskjernen Ex: IP (v4 og v6), ICMP, RIP, OSPF, BGP,...
Datalinjelaget	Ramme (Frame)	(Pålitelig) Levering av ramme fra nabo-node til nabo-node Ex: Ethernet II, FDDI, IEEE 802.11 ...
Fysisk	Bit	(Kode og) Flytte enkeltbit mellom kommunikasjonspartnere. Ex: 10BaseT, ...

- Sender: Each layer takes data from the layer above
 - Adds information (header), creates new data unit
 - Delivers new data to the layer below
- Receiver processes data in reverse order



HTTP



N. Nettavisen

nettavisen.no

Fullfør oppdateringen



SOGNDAL - VÅLERENGA

FOSSHAUGANE CAMPUS

URDAG 9. NOV.

KL. 16.00

Sparebanken West | OSLO TIL LIGAEN



2020 2022 2024

Tror festen er over: Brutal kalddusj til norske sparere



**Dennis (25) tok et
fatalt valg: – Uvirkelig**



Shoppingtips med annonselenker

**Gavene hun
virkelig ønsker seg**

ANNONSE



SOGNDAL - VÅLERENGA

FOSSHAUGANE CAMPUS

LAURDAG 9. NOV.

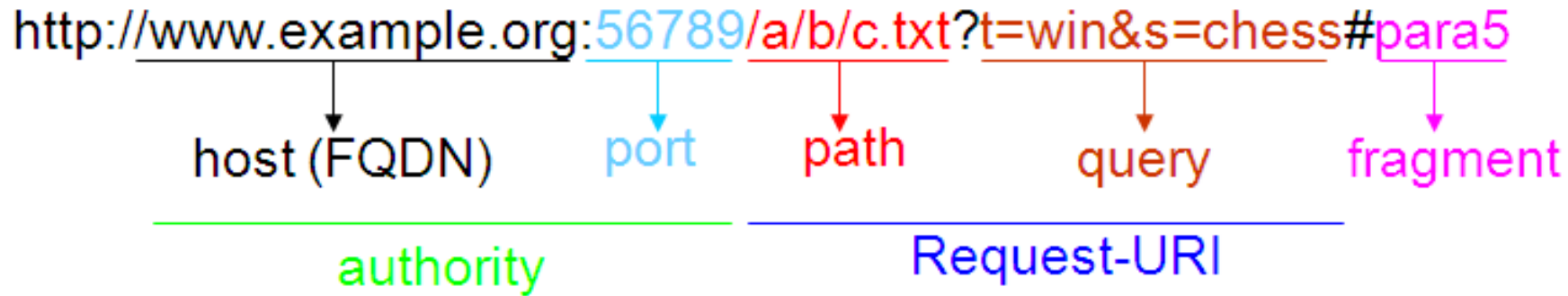
KL. 16.00

Sparebanken West | OSLO TIL LIGAEN

Web expression

- Web page
 - Consists of "objects", addressed by a URI
- Typically, the website has
 - A base HTML page (index.html), multiple object references
- The URI (url) consists of
 - protocol://user:password@hostname:port/filepath/filename#anchor?parameters
(protocol:// user:pwd@host:port/ path ?parameters #anchor)
 - http://test.bengt.no/webseite/ressursnavn.txt
- The user agent on the web is a **browser**.
 - Chrome, Internet Explorer, Mozilla, ...
- The server on the web is called a **web server**.
 - Apache, MS IIS, Nginx, ...

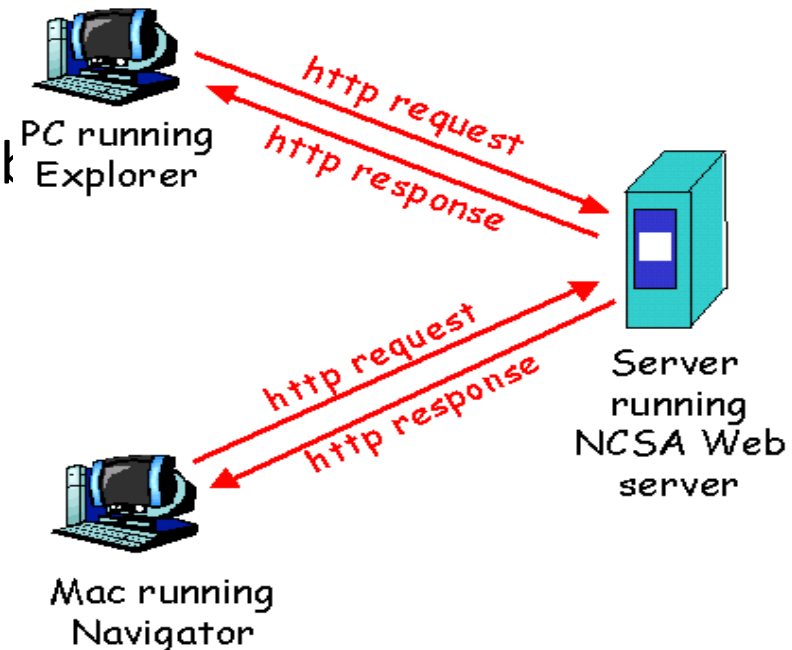
HTTP URL



- The browser performs a DNS lookup and establishes a TCP connection to the “authority”.
- Then follows the "file path" on the server (resource ID)
- After ? follows arguments to script/program
- After # typically an **anchor** / position within a resource (“document”) (< **a** href=)

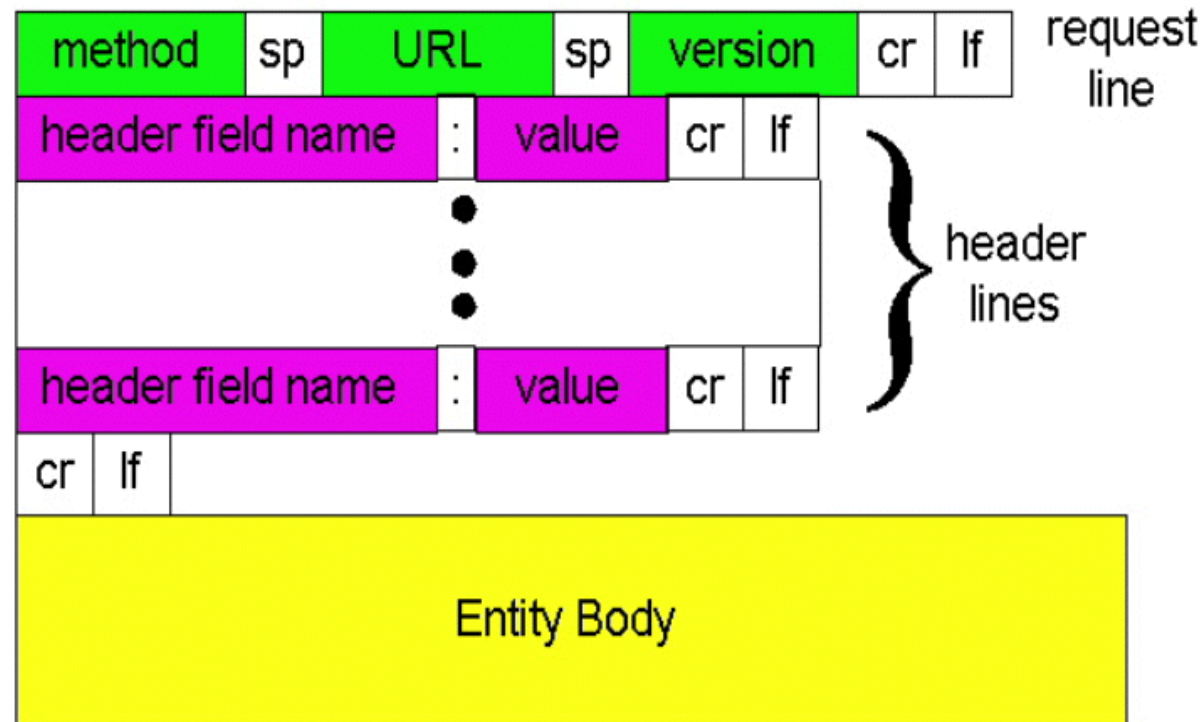
HTTP (HyperText Transfer Protocol)

- The Web's application protocol
- Client/server model
 - The client requests, receives, and displays objects
 - The server sends objects on demand
- Using **TCP** transport service
 - Client sets up the TCP connection
 - The server accepts the connection from the
 - Client and server exchange information
 - The TCP connection is closed.
- *does not* store any information about previous requests (**stateless**)

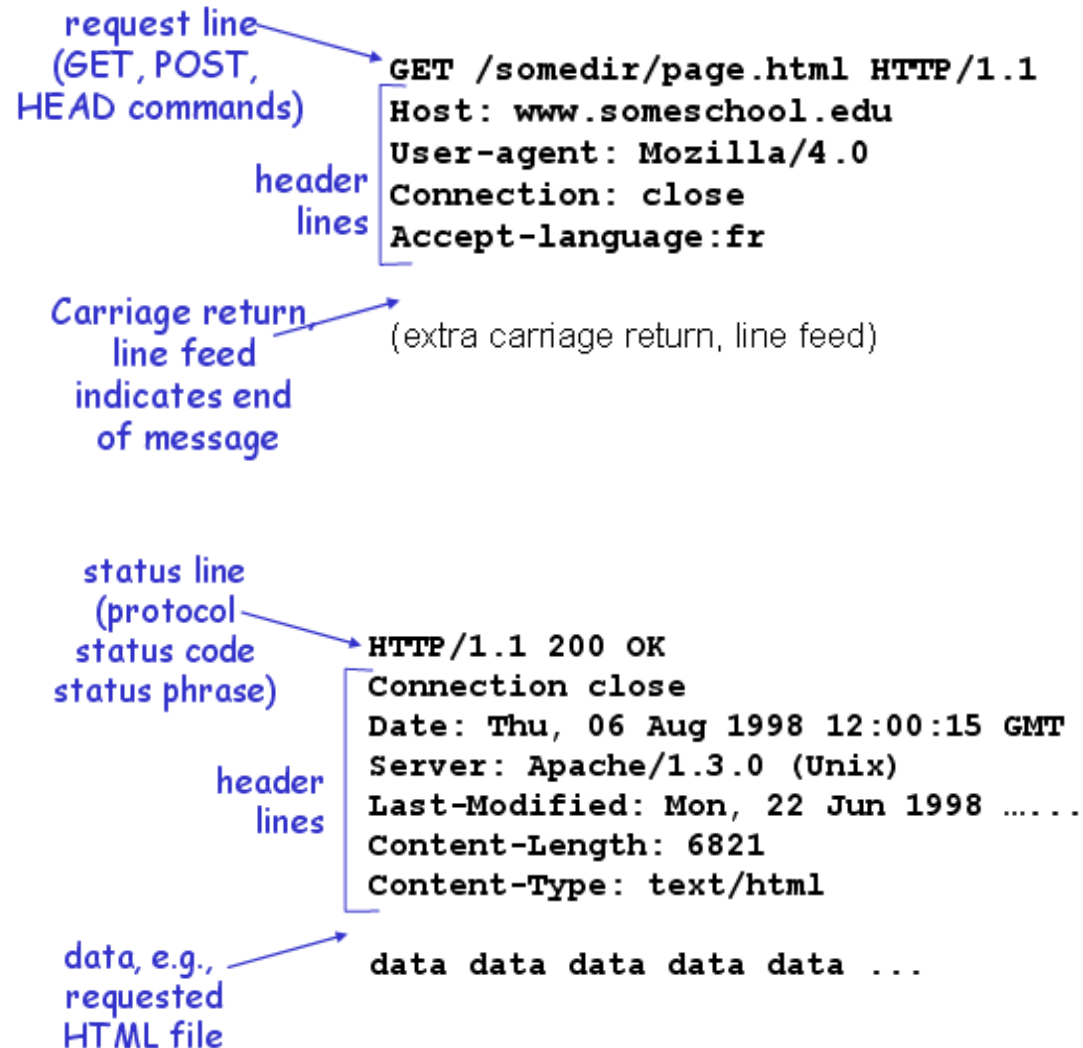


HTTP message format

Message **headers** are encoded in **7 bit ASCII** format.



HTTP message format



HTTP response status codes

- Located in the first line of the answer
- Examples:

200 OK

- query successful, object comes later in message

301 Moved Permanently

- requested object moved, new address later in the message

400 Bad Request

- query not understood by the server

404 Not Found

- requested document/file not found on this server

505 HTTP Version Not Supported

System:

- Three-digit status code

1 xx = Informational

2 xx = Success

3 xx = Redirection

(alternate URL is supplied)

4 xx = Client Error

5 xx = Server Error

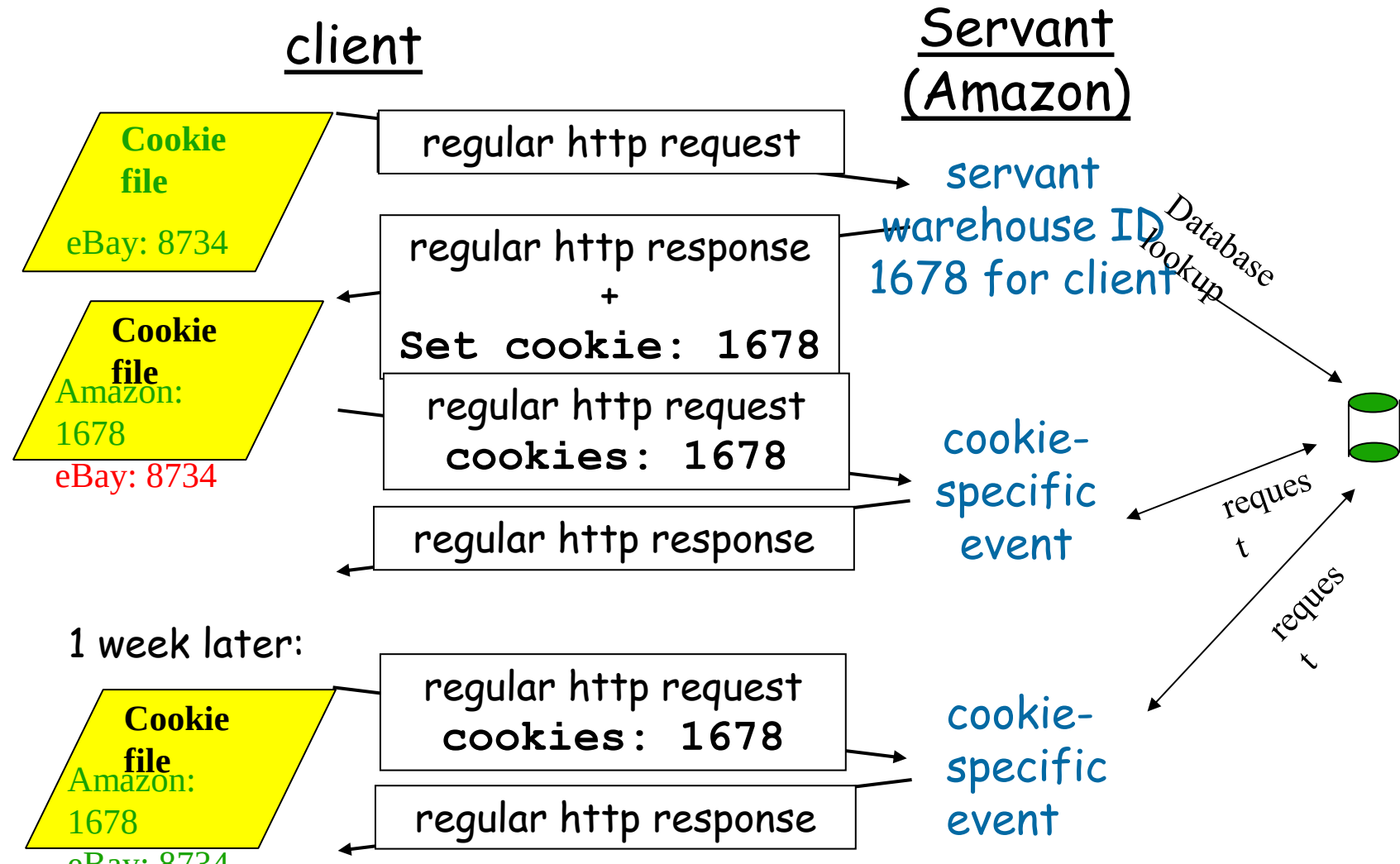
418 I'm a Teapot

Status response code indicates that the server refuses to brew coffee because it is, permanently, a teapot... [standardised in RFC 1 April 1998]

Preserve state with cookie

Since HTTP is not stateful, "modern" web applications will not work; for example, online stores.

Cookie keeps state over HTTP.



```
www.eastwillsecurity.com - PuTTY
GET / HTTP/1.1
Host: www.eastwillsecurity.com

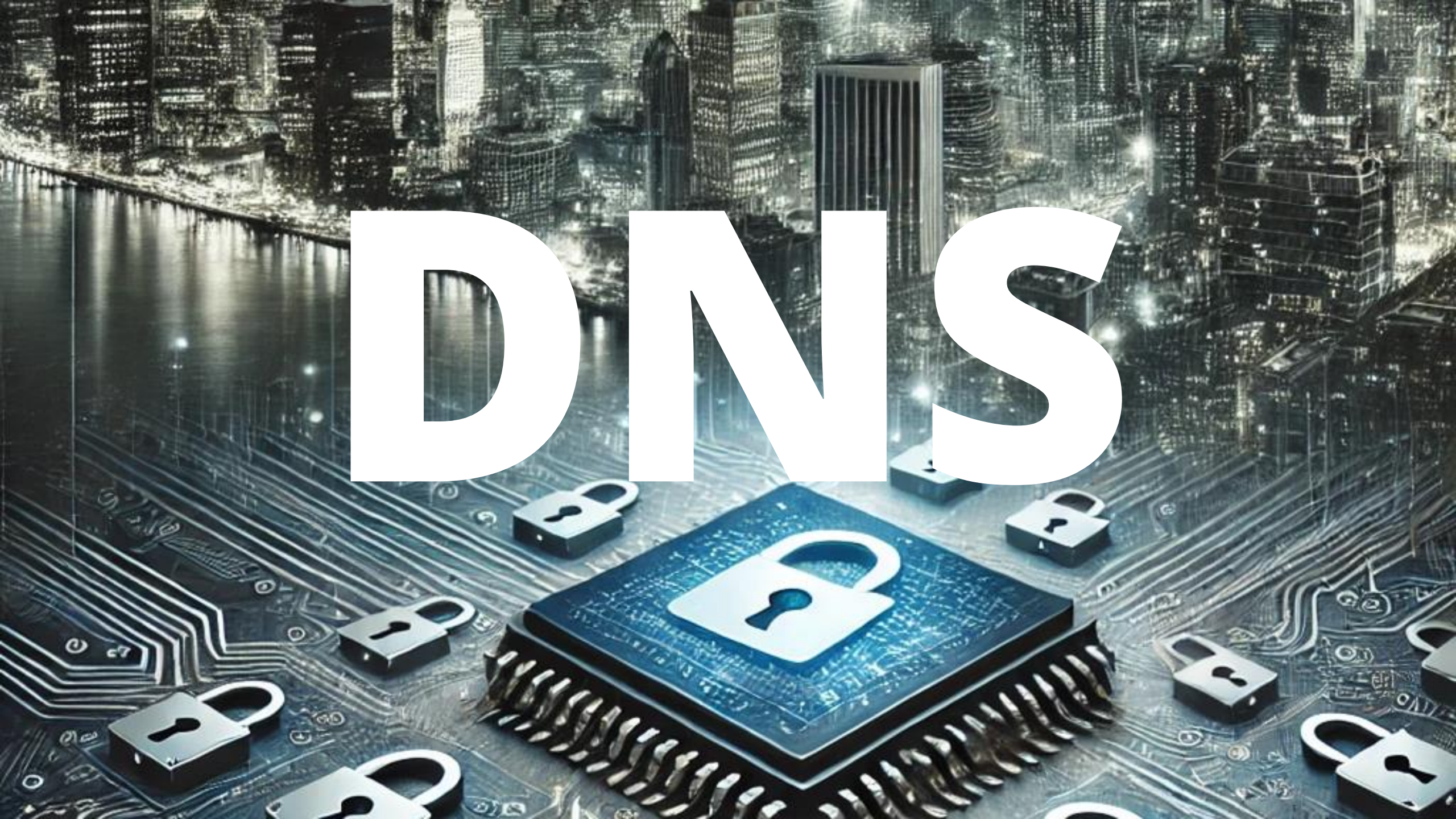
HTTP/1.1 200 OK
Date: Tue, 05 Nov 2024 07:49:11 GMT
Server: Apache
Last-Modified: Tue, 05 Nov 2024 07:48:38 GMT
Vary: Accept-Encoding
Content-Type: text/html
X-Onecom-Cluster-Name:
X-Varnish: 8079511 200871719
Age: 82
Via: 1.1 webcache1 (Varnish/trunk)
ETag: W/"14f-62625a111d8eb-gzip"
Accept-Ranges: bytes
Content-Length: 335
Connection: keep-alive

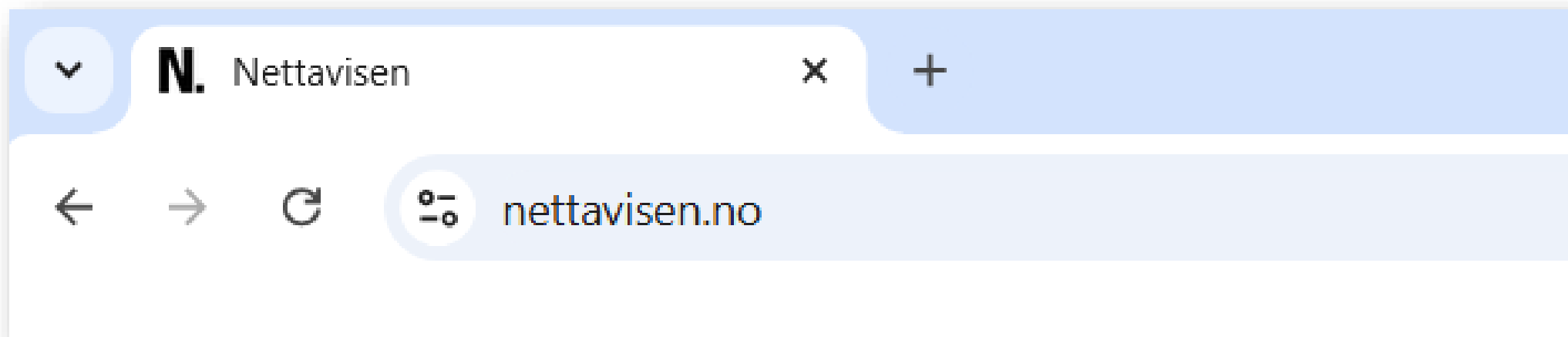
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Eastwill Testpage</title>
</head>
<body>
  <h1>Eastwill Testpage</h1>
  <p>This is a standard and simple webpage, this host serves the same contents
over both HTTP and HTTPS.</p>
  
</body>
</html>
```

Because commands in the HTTP protocol are in 7-bit ASCII, you can send manual requests to download a web page – although most people will still prefer to view the web page in a graphical version. 😊



DNS



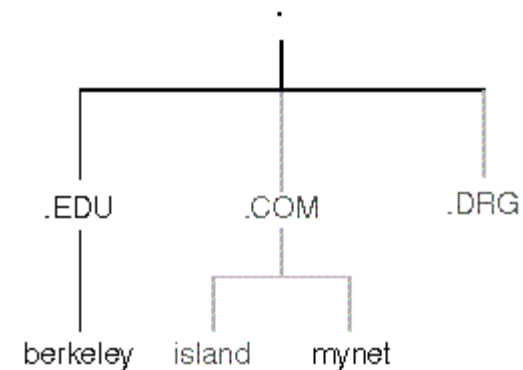
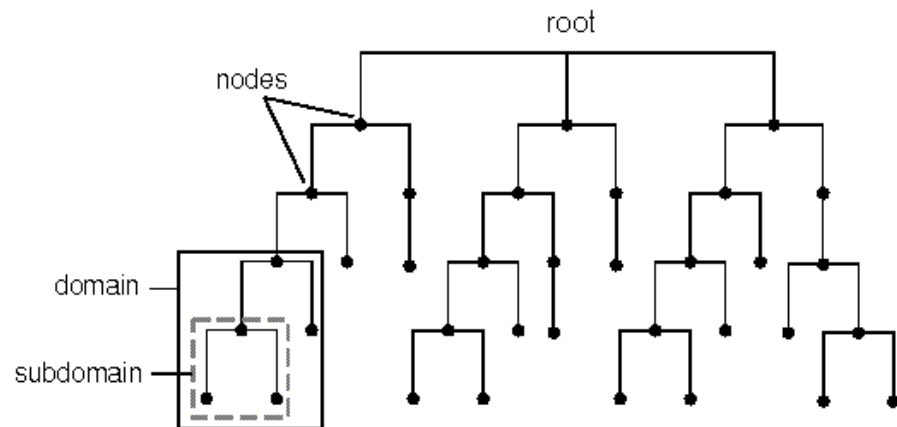


DNS (Domain Name System)

- People
 - Name, social security number
 - "Ola Nordmann", 161165 42796
- Internet
 - IP address, Name
 - 10.11.1.21, www.kristinia.no
- Resolving this on the Internet with DNS
 - Distributed database on many name servers
 - Application layer protocol for associating names and IP addresses

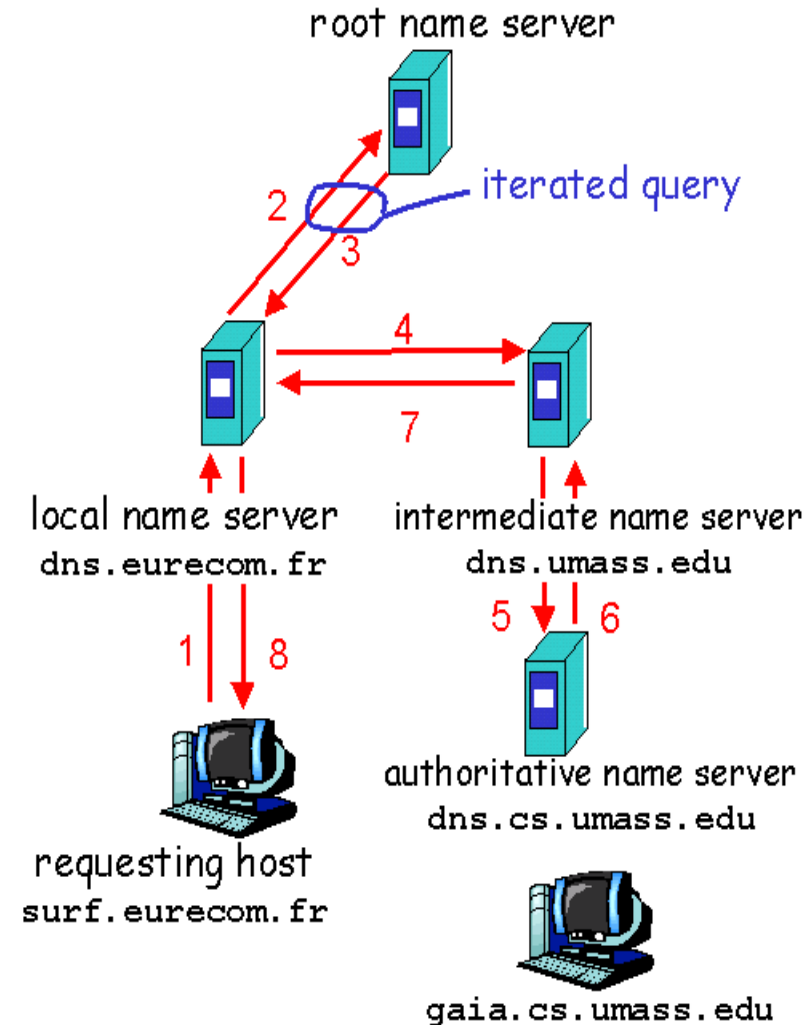
DNS name servers

- Not centralized!
 - Prevent the entire network from going down with the name server
 - Avoid traffic congestion
 - Centralized database is always “far” away
 - Scalable
- Name servers are distributed hierarchically



Repeated queries

- Typically, queries are **recursive**
 - A queries on behalf of B and returns the answer to B to the local (authoritative) name server
 - Typical from user
- Queries can also be **iterative**
 - A queries on behalf of B and returns the next server's address to A, who then queries it himself
 - typically from local name servers to root, TLD (Top Level Domain) and other local name servers



Caching and updating

- Name servers cache DNS mappings
- Caching disappears after a while (TTL timeout)
- The DNS resolver on your own OS also caches..
- Mechanisms for enrollment and updating are under development at the IETF (Internet Engineering Task Force)
 - Dynamic update
 - Security
 - mmm

```
C:\>ipconfig /flushdns
Windows IP Configuration
Successfully flushed the DNS Resolver Cache.
C:\>ipconfig /displaydns
Windows IP Configuration

www.google.no
-----
Record Name . . . . . : www.google.no
Record Type . . . . . : 5
Time To Live . . . . . : 73
Data Length . . . . . : 8
Section . . . . . : Answer
CNAME Record . . . . . : www.google.com

maps.google.no
-----
Record Name . . . . . : maps.google.no
Record Type . . . . . : 5
Time To Live . . . . . : 31
Data Length . . . . . : 8
Section . . . . . : Answer
CNAME Record . . . . . : maps.google.com
```

Norwegian letters?

- To support character sets other than ASCII, DNS now uses *Punycode* (Internationalized Domain Names – IDN)
- `ø1.no` is in the zone file noted `xn--l-4ga.no` :

```
xn--l-4ga.no
-----
Record Name . . . . . : xn--l-4ga.no
Record Type . . . . . : 1
Time To Live . . . . . : 71852
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 83.143.81.86
```

- **Punycode** is defined in RFC 3492
- For example `www.østby.net` = `www.xn--stby-fra.net`

Administrator: Command Prompt - nslookup

C:\Windows\system32>nslookup

Default Server: UnKnown

Address: 192.168.89.247

> set type=NS

> no

Server: UnKnown

Address: 192.168.89.247

Non-authoritative answer:

no nameserver = z.nic.no

no nameserver = y.nic.no

no nameserver = not.norid.no

no nameserver = njet.norid.no

no nameserver = x.nic.no

no nameserver = i.nic.no

> eos-utvalget.no

Server: UnKnown

Address: 192.168.89.247

Non-authoritative answer:

eos-utvalget.no nameserver = ns2.hyp.net

eos-utvalget.no nameserver = ns3.hyp.net

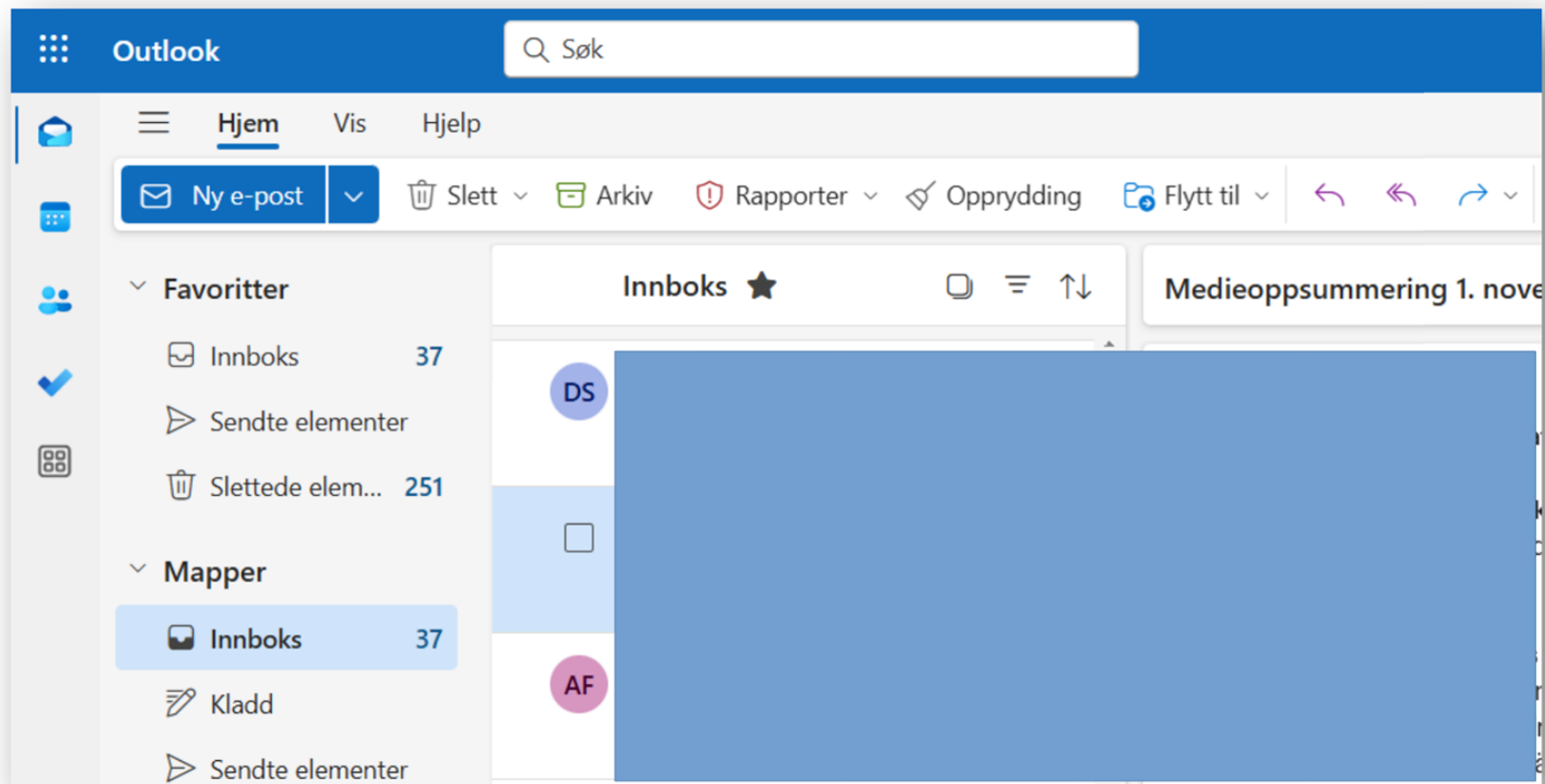
eos-utvalget.no nameserver = ns1.hyp.net

> _

The background is a composite image. The top half shows a night view of a city skyline with many lit-up skyscrapers and a body of water in the foreground. The bottom half shows a close-up of a blue circuit board with various electronic components. Several silver padlocks are scattered across the circuit board, with one large padlock prominently placed in the center. The letters 'S', 'M', and 'T' are overlaid in white on the top half of the image, and the letter 'D' is overlaid in white on the bottom half, partially covering the circuit board and padlocks.

SMT

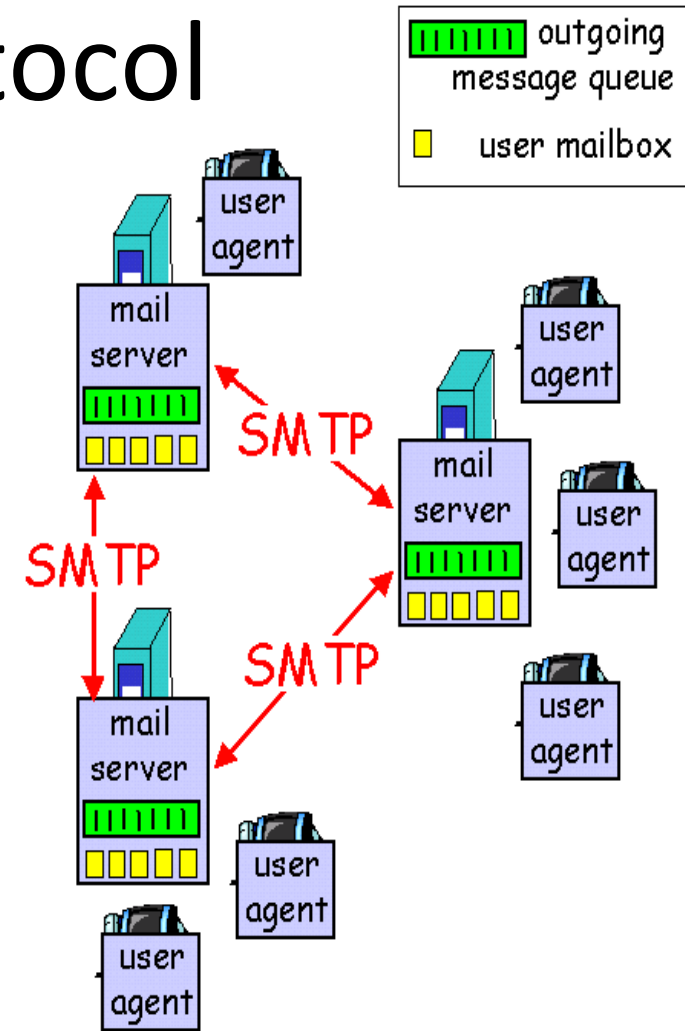
D



Simple Mail Transfer Protocol

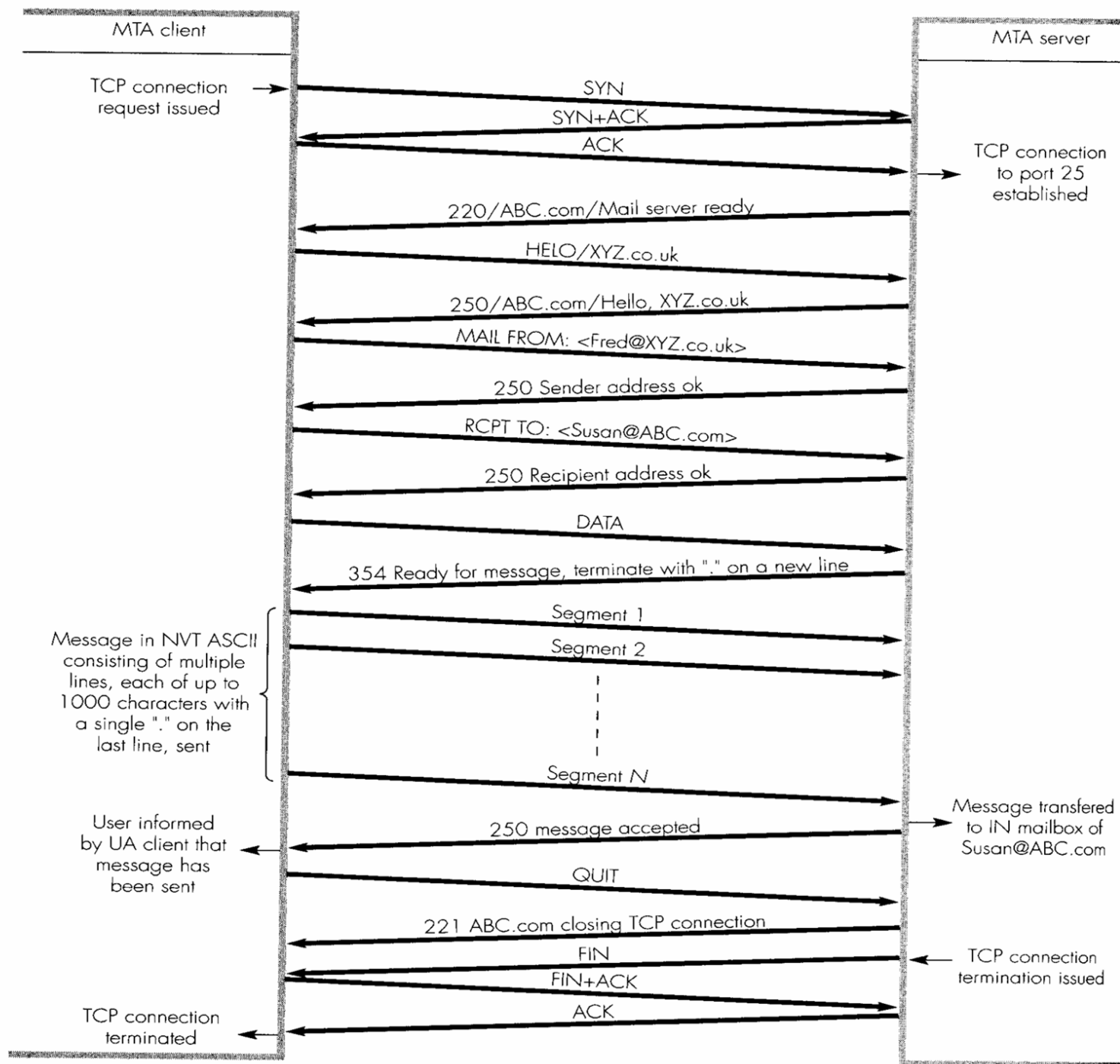
- Mailbox (house) containing unread mail for the user
- Queue for mail to be sent
- SMTP protocol between servers for **sending** mail
- The client acts as a sender server to the SMTP server it is set up to use.

To receive email, use e.g. POP3 or IMAP.

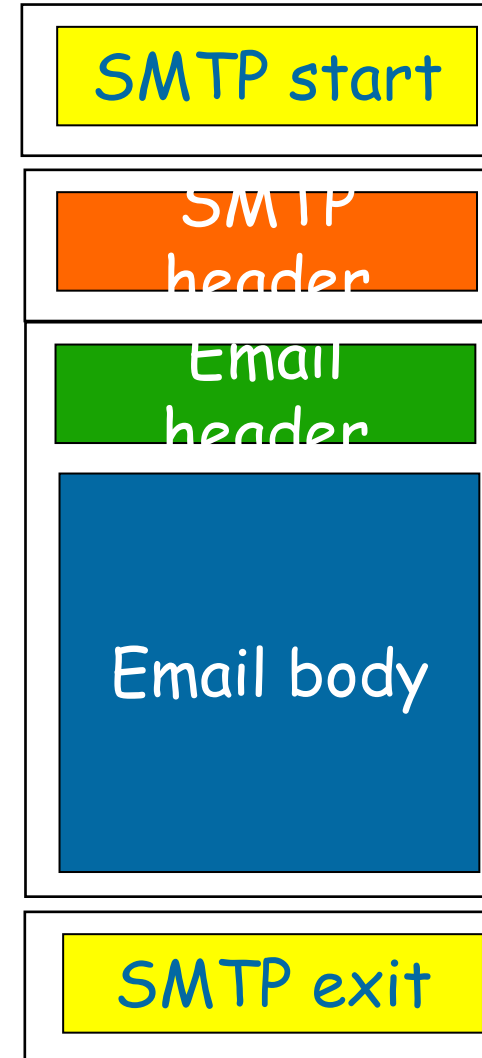


Simple Mail Transfer Protocol

- Uses TCP to transfer mail from client to server, port 25
- Direct transfer from server to server
- 3 transfer phases
 - Handshake
 - Transmission
 - Termination
- Transfer in ASCII text
 - Commands and status codes
- The message part is in 7-bit ASCII



- SMTP conversation
 - HELLO
- SMTP header
 - MAIL FROM:
 - RCPT TO:
 - DATA
- Email header
 - From:
 - To:
 - Subject:
 - These are **not SMTP** commands!
- Blank line (Two CRLF)
- Body
 - Only 7 bit ASCII text
 - Sent with a period at the start of a line followed by a line break
- QUIT



SMTP vs HTTP

- **SMTP** uses persistent connection
- Some character strings are illegal in messages
- Everything is in ASCII code.
 - Among other things, the message is therefore recoded (base64, Uuencode, hex64 ...)
 - CR LF . CR LF ends a message
- **HTTP** pulls data, email pushes data
- **HTTP** (usually) transfers one object per message
- **SMTP** can transfer many (recoded to ASCII) objects per single message ("attachments")

Because commands in the SMTP protocol – like HTTP – are in 7-bit ASCII, you can send manual requests to send an email – although most people will probably still prefer to use a graphical version. 😊

Email (my domain): **test@eastwillsecurity.com**

Password (for testing purposes only): **test2pwd48sett**

Domain host: one.com, email server: send.one.com

Both of these need to be encoded:

dGVzdEBIYXN0d2lsbHNlY3VyaXR5LmNvbQ==

dGVzdDJwd2Q0OHNIHQ=

```
EHLO eastwillsecurity.com
```

```
AUTHORIZED LOGIN
```

```
dGVzdEBIYXN0d2lsbHNlY3VyaXR5LmNvbQ==
```

```
dGVzdDJwd2Q0OHNIHQ=
```

```
MAIL FROM:< test@eastwillsecurity.com >
```

```
BODY=8BITMIME
```

```
RCPT TO: < bengt.ostby@not-a-real.no >
```

```
DATA
```

```
Date: Thu, 5 November 2024 12:15:29 +0100
```

```
From: Bengt < test@eastwillsecurity.com >
```

```
Subject: Test
```

```
To: bengt.ostby@not-a-real.no
```

```
Test email, here you can write whatever you want
```

```
.
```

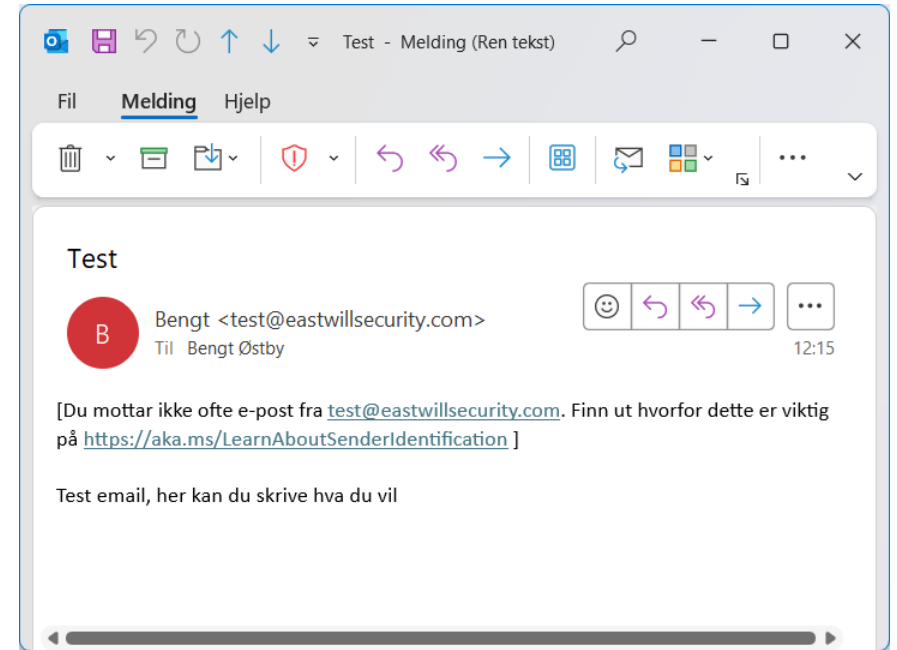
```
QUIT
```

PuTTY (inactive)

```
220 mailrelay3.pub.mailoutpod3-cph3.one.com ESMTP
EHLO eastwillsecurity.com
250-mailrelay3.pub.mailoutpod3-cph3.one.com
250-PIPELINING
250-CHUNKING
250-SIZE 104857600
250-STARTTLS
250-AUTH LOGIN PLAIN
250-ENHANCEDSTATUSCODES
250 8BITMIME
AUTH LOGIN
334 VXNlcm5hbWU6
dGVzdEB1YXN0d2lsbHN1Y3VyaXR5LmNvbQ==
334 UGFzc3dvcmQ6
dGVzdDJwd2Q0OHNldHQ=
235 2.7.0 Ok
MAIL FROM:<test@eastwillsecurity.com> BODY=8BITMIME
250 2.1.0 Ok
RCPT TO:<bengt.ostby@.no>
250 2.1.5 Ok
DATA
354 End data with <CR><LF>.<CR><LF>
Date: Thu, 5 November 2024 12:15:29 +0100
From: Bengt <test@eastwillsecurity.com>
Subject: Test
To: bengt.ostby@.no

Test email, her kan du skrive hva du vil

.
250 2.0.0 Ok: queued as fd49d34d-9b54-11ef-8891-4bfe3c5334f1
QUIT
221 2.0.0 Bye
```



FTP




```
Administrator: Command Prompt - ftp ftp.ed.ac.uk

D:\_Internt>ftp ftp.ed.ac.uk
Connected to ftp1.is.ed.ac.uk.
220-University of Edinburgh Anonymous FTP server
=====

When requested for a username enter 'ftp' or 'anonymous'. If you have
problems, try using a dash (-) as the first character of your password.
If you still have problems or wish to make a comment then send email to
```

FileZilla

Fil Rediger Visning Overfør Server Bokmerker Hjelp

Server: home.nith.no Brukernavn: blistog Passord: Port: 22 Hurtigtilkobling

Status: Kobler til home.nith.no...
Respons: fzSftp started
Kommando: open "blistog@home.nith.no" 22
Kommando: Pass: *****
Status: Connected to home.nith.no
Status: Mottar mappeliste...
Kommando: pwd
Respons: Current directory is: "/home/blistog"

Lokalt sted: C:\Users\blistog\DB110-Ovinger\

+	..
+	.zenmap
+	_eric5
+	AppData
+	Application Data
+	Contacts
+	Cookies

Filnavn	Filstørrelse	Filtype	Sist modifisert
..			
Oving_4_V2011.sql	4 683	SQL File	15.02.2011 12:21:40

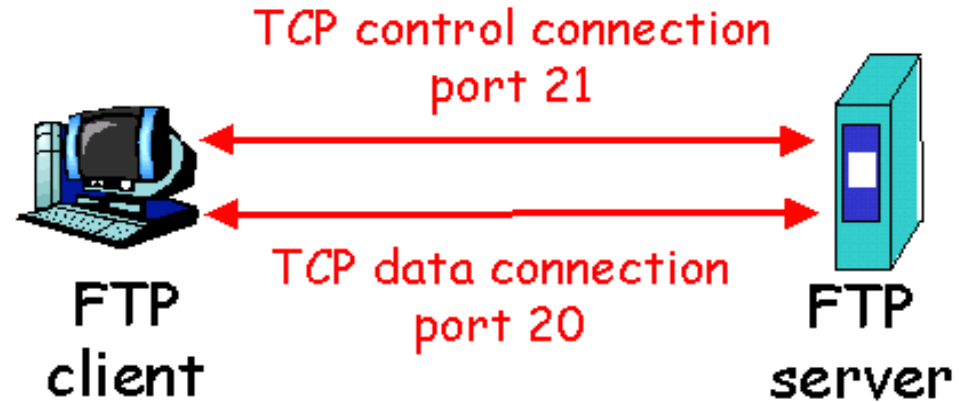
Server: /home/blistog

?	/
?	home
+	blistog

Filnavn	Filstørrelse	Filtype	Sist modifisert	Rettigheter
pics		File folder	15.09.2012 15:4...	drwxr-xr-x
public_html		File folder	10.04.2012	drwxr-xr-x
py		File folder	27.05.2011	drwxr-xr-x
RF300		File folder	27.05.2011	drwxr-xr-x
svn1		File folder	18.01.2012	drwxr-xr-x
TEST		File folder	27.05.2011	drwx-----

File Transfer Protocol

- Transferring files
- Uses client/server model
- Fast transfer
- *Can* use browser as interface



- Two parallel connections are opened
 - Control (port 21): Transmits commands and responses
 - Data (20): Transmitting data
- FTP is not stateless
 - Client and servant have a **shared** "understanding" of what should be done in what order, and how far one has come.
 - "Remembers" e.g. ID/password and valid folder

File Transfer Protocol

- Sent as ASCII text
- Commands
 - USER *username*, PASS *password*
 - LIST gives a list of files in the directory
 - RETR *filename* , STOR *filename* retrieves/stores file
- Return codes are similar to HTTP
 - 125 Data connection already open; transfer starting
 - 331 Username OK, password required
 - 425 Can't open data connection
 - 452 Error writing file

Security, HTTP and FTP

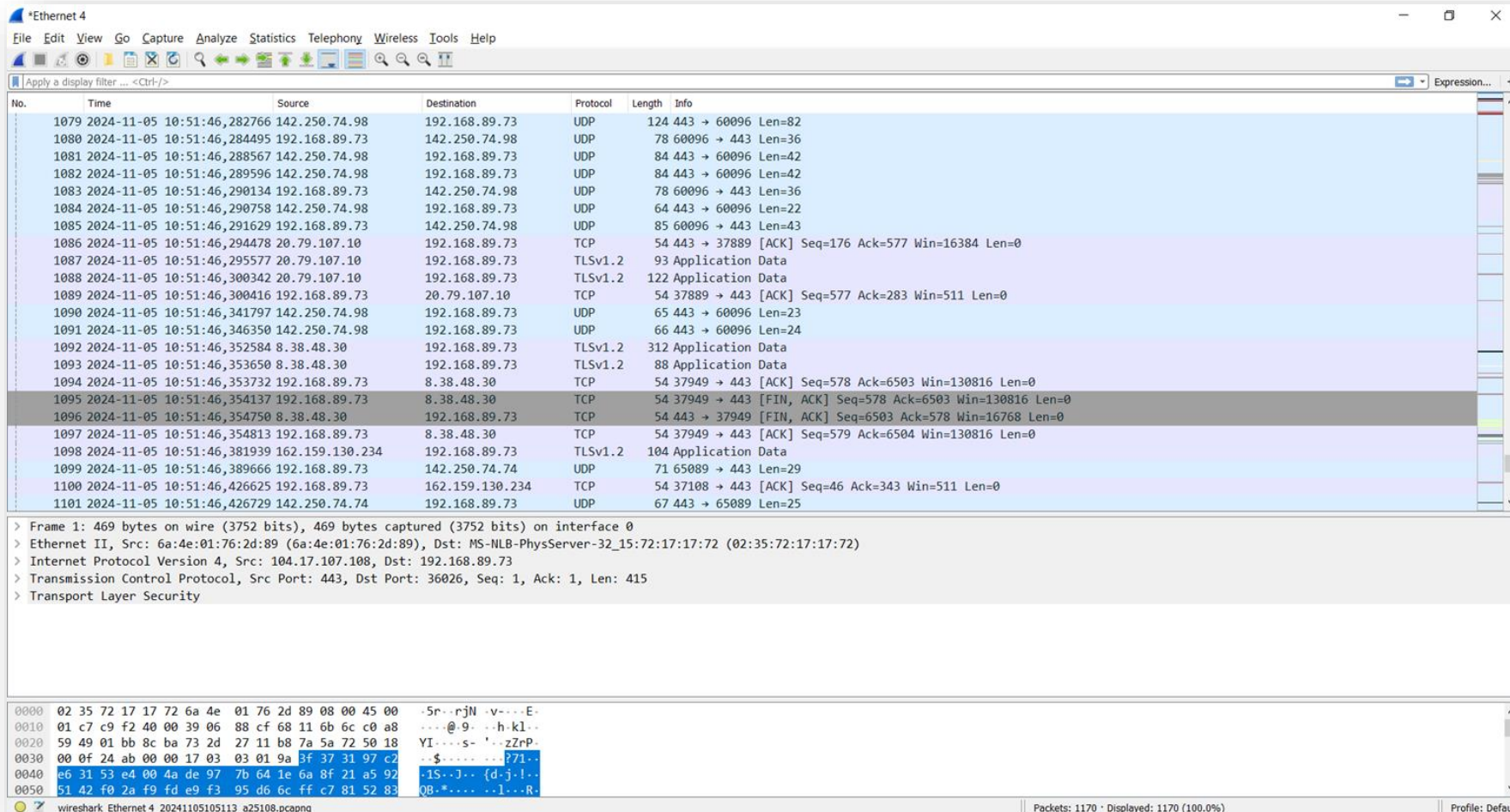
- All protocols we have looked at are UNENCRYPTED, data in clear text!
- SFTP is not a new version of FTP, but a completely new protocol
 - Typically, it uses SSH (Secure Shell) to establish an encrypted connection to an **sftp file server**.
 - Standard port 22
- HTTP S is regular HTTP over an encrypted transport layer connection
 - typically uses the TLS/SSL protocols to create an encrypted TCP connection
 - Default port 443 .

Wireshark



Because all the "basic" protocols in networks and over the internet are text-based - and sent as 7-bit ASCII - one can easily intercept and interpret data by looking at a raw packet stream from a computer - a widely used tool for this is Wireshark. Wireshark saves files in PCAP format.

Wireshark



Wireshark

No.	Time	Source	Destination	Protocol	Length	Info
221	2024-11-05 10:56:10.429070	192.168.89.73	77.111.240.75	TCP	54	37982 → 80 [ACK] Seq=566 Ack=361 Win=512 Len=0
1103	2024-11-05 10:56:37.745925	192.168.89.73	77.111.240.75	TCP	54	37981 → 80 [FIN, ACK] Seq=2 Ack=1 Win=515 Len=0
1111	2024-11-05 10:56:37.781399	77.111.240.75	192.168.89.73	TCP	54	80 → 37981 [FIN, ACK] Seq=1 Ack=3 Win=502 Len=0
1112	2024-11-05 10:56:37.781605	192.168.89.73	77.111.240.75	TCP	54	37981 → 80 [ACK] Seq=3 Ack=2 Win=515 Len=0
2482	2024-11-05 10:56:44.694702	192.168.89.73	77.111.240.75	HTTP	528	GET / HTTP/1.1
2483	2024-11-05 10:56:44.696868	192.168.89.73	77.111.240.75	TCP	66	38001 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
2484	2024-11-05 10:56:44.739870	77.111.240.75	192.168.89.73	TCP	66	80 → 38001 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1360 SACK_PERM=1 WS=128
2485	2024-11-05 10:56:44.740112	192.168.89.73	77.111.240.75	TCP	54	38001 → 80 [ACK] Seq=1 Ack=1 Win=131840 Len=0
2486	2024-11-05 10:56:44.744556	77.111.240.75	192.168.89.73	HTTP	682	HTTP/1.1 200 OK (text/html)
2487	2024-11-05 10:56:44.762330	192.168.89.73	77.111.240.75	HTTP	455	GET /ew_logo.png HTTP/1.1
2488	2024-11-05 10:56:44.823702	77.111.240.75	192.168.89.73	TCP	1414	80 → 37982 [ACK] Seq=989 Ack=1441 Win=501 Len=1360 [TCP segment of a reassembled PDU]
2489	2024-11-05 10:56:44.824768	77.111.240.75	192.168.89.73	TCP	1414	80 → 37982 [PSH, ACK] Seq=2349 Ack=1441 Win=501 Len=1360 [TCP segment of a reassembled PDU]
2490	2024-11-05 10:56:44.824864	192.168.89.73	77.111.240.75	TCP	54	37982 → 80 [ACK] Seq=1441 Ack=3709 Win=515 Len=0
2491	2024-11-05 10:56:44.825885	77.111.240.75	192.168.89.73	TCP	1414	80 → 37982 [ACK] Seq=3709 Ack=1441 Win=501 Len=1360 [TCP segment of a reassembled PDU]
2492	2024-11-05 10:56:44.826972	77.111.240.75	192.168.89.73	TCP	1414	80 → 37982 [PSH, ACK] Seq=5069 Ack=1441 Win=501 Len=1360 [TCP segment of a reassembled PDU]
2493	2024-11-05 10:56:44.827052	192.168.89.73	77.111.240.75	TCP	54	37982 → 80 [ACK] Seq=1441 Ack=6429 Win=515 Len=0
2494	2024-11-05 10:56:44.828087	77.111.240.75	192.168.89.73	TCP	1414	80 → 37982 [ACK] Seq=6429 Ack=1441 Win=501 Len=1360 [TCP segment of a reassembled PDU]
2495	2024-11-05 10:56:44.829140	77.111.240.75	192.168.89.73	TCP	1414	80 → 37982 [PSH, ACK] Seq=7789 Ack=1441 Win=501 Len=1360 [TCP segment of a reassembled PDU]
2496	2024-11-05 10:56:44.829218	192.168.89.73	77.111.240.75	TCP	54	37982 → 80 [ACK] Seq=1441 Ack=9149 Win=515 Len=0
2497	2024-11-05 10:56:44.830147	77.111.240.75	192.168.89.73	TCP	1414	80 → 37982 [ACK] Seq=9149 Ack=1441 Win=501 Len=1360 [TCP segment of a reassembled PDU]
2498	2024-11-05 10:56:44.831287	77.111.240.75	192.168.89.73	TCP	1414	80 → 37982 [PSH, ACK] Seq=10509 Ack=1441 Win=501 Len=1360 [TCP segment of a reassembled PDU]
2499	2024-11-05 10:56:44.831389	192.168.89.73	77.111.240.75	TCP	54	37982 → 80 [ACK] Seq=1441 Ack=11869 Win=515 Len=0
2500	2024-11-05 10:56:44.832503	77.111.240.75	192.168.89.73	TCP	1414	80 → 37982 [ACK] Seq=11869 Ack=1441 Win=501 Len=1360 [TCP segment of a reassembled PDU]

Wireshark

2482	2024-11-05 10:56:44,694702	192.168.89.73	77.111.240.75	HTTP	528 GET / HTTP/1.1
2483	2024-11-05 10:56:44,696868	192.168.89.73	77.111.240.75	TCP	66 38001 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
2484	2024-11-05 10:56:44,739870	77.111.240.75	192.168.89.73	TCP	66 80 → 38001 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1360 SACK_PERM=1 WS=128
2485	2024-11-05 10:56:44,740112	192.168.89.73	77.111.240.75	TCP	54 38001 → 80 [ACK] Seq=1 Ack=1 Win=131840 Len=0
2486	2024-11-05 10:56:44,744556	77.111.240.75	192.168.89.73	HTTP	682 HTTP/1.1 200 OK (text/html)

> Frame 2486: 682 bytes on wire (5456 bits), 682 bytes captured (5456 bits) on interface 0

▼ Ethernet II, Src: 6a:4e:01:76:2d:89 (6a:4e:01:76:2d:89), Dst: MS-NLB-PhysServer-32_15:72:17:17:72 (02:35:72:17:17:72)

▼ Destination: MS-NLB-PhysServer-32_15:72:17:17:72 (02:35:72:17:17:72)

Address: MS-NLB-PhysServer-32_15:72:17:17:72 (02:35:72:17:17:72)

.... 1. = LG bit: Locally administered address (this is NOT the factory default)

.... 0 = IG bit: Individual address (unicast)

▼ Source: 6a:4e:01:76:2d:89 (6a:4e:01:76:2d:89)

Address: 6a:4e:01:76:2d:89 (6a:4e:01:76:2d:89)

.... 1. = LG bit: Locally administered address (this is NOT the factory default)

.... 0 = IG bit: Individual address (unicast)

Type: IPv4 (0x0800)

▼ Internet Protocol Version 4, Src: 77.111.240.75, Dst: 192.168.89.73

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 668
Identification: 0xa2a8 (41640)

> Flags: 0x4000, Don't fragment

Time to live: 54
Protocol: TCP (6)
Header checksum: 0x4807 [validation disabled]
[Header checksum status: Unverified]
Source: 77.111.240.75
Destination: 192.168.89.73

0010	02 9c a2 a8 40 00 36 06	48 07 4d 6f f0 4b c0 a8	@-6- H-Mo-K-.
0020	59 49 00 50 94 5e f5 1b	8b cb 8f 71 69 d1 50 18	YI-P-^...-qi-P-
0030	01 f5 1c 46 00 00 48 54	54 50 2f 31 2e 31 20 32	...F-HT TP/1.1 2
0040	30 30 20 4f 4b 0d 0a 44	61 74 65 3a 20 54 75 65	00 OK-D ate: Tue

Wireshark

```
▼ Hypertext Transfer Protocol
  > HTTP/1.1 200 OK\r\n
    Date: Tue, 05 Nov 2024 09:04:07 GMT\r\n
    Server: Apache\r\n
    Last-Modified: Tue, 05 Nov 2024 07:48:38 GMT\r\n
    ETag: "14f-62625a111d8eb-gzip"\r\n
    Vary: Accept-Encoding\r\n
    Content-Encoding: gzip\r\n
  > Content-Length: 235\r\n
    Content-Type: text/html\r\n

▼ Line-based text data: text/html (12 lines)
  <!DOCTYPE html>\r\n
  <html lang="en">\r\n
  <head>\r\n
    <meta charset="UTF-8">\r\n
    <title>Eastwill Testpage</title>\r\n
  </head>\r\n
  <body>\r\n
    <h1>Eastwill Testpage</h1>\r\n
    <p>This is a standard and simple webpage, this host serves the same contents over both HTTP and HTTPS.</p>\r\n
    \r\n
  </body>\r\n
</html>
```



← → ↻ ⚠ Ikke sikker eastwillsecurity.com

Eastwill Testpage

This is a standard and simple webpage, this host serves the same contents over both HTTP and HTTPS.



...
Eastwill Testpage
← → ↻

Eastwill Testpage

This is a standard and s

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Transport Layer Security

TLSv1.3 Record Layer: Application Data Protocol: http-over-tls

Opaque Type: Application Data (23)

Version: TLS 1.2 (0x0303)

Length: 282

Encrypted Application Data: abd6bdbea07163312ebb86a4129b6ff83e9c6dfd4d18db36...

0000	02 35 72 17 17 72 6a 4e 01 76 2d 89 08 00 45 00	-5r..rjN -v....E.
0010	01 47 13 57 40 00 36 06 d8 ad 4d 6f f0 4b c0 a8	-G.W@-6- ..Mo-K..
0020	59 49 01 bb 94 7c d6 1f 59 d3 ca 6f da ce 50 18	YI.... .. Y..o..P.
0030	01 f5 23 d0 00 00 17 03 03 01 1a ab d6 bd be a0	..#.....
0040	71 63 31 2e bb 86 a4 12 9b 6f f8 3e 9c 6d fd 4d	qc1..... .o>.m.M
0050	18 db 36 e2 68 57 96 2f ea 32 1e e3 be dc e5 ef	..6.hW./ .2.....
0060	ae e2 55 0f 39 47 2a 54 62 22 a2 e0 7f 91 e0 14	..U.9G*T b".....
0070	71 98 f2 dc 9c e5 3d c9 b9 48 15 80 a6 dc 22 f8	q.....=. .H....".
0080	32 0f 0f 6d 67 ae b0 31 50 cf c6 0b e9 b5 4d 00	2..mg..1 P.....M.
0090	12 30 eb 6f 93 dd ff b6 0a 3e f6 86 8f e9 1e 8f	.0.o.... .>.....
00a0	c1 76 59 1d dc f8 01 5b bf 66 9e 36 fe 2a c0 5f	.vY....[.f.6.*._
00b0	dc 4f 0c 26 0f 0c 54 d6 43 65 c3 e8 4c 9e 50 8e	.0.&..T. Ce..L.P.
00c0	c3 73 76 86 ed 0e a6 d6 cc 13 3a 8d 35 3e 62 ed	.sv..... ..:5>b.
00d0	1c 60 14 df 59 17 11 d8 78 c7 6f b7 f4 f3 26 34	.~..Y... x.o...&4
00e0	6a 7e e9 66 2b 19 4c 7b e6 bb aa 7c 7c 37 fb 7e	j~.f+.L{ ... 7~
00f0	be 52 07 4b 8f 38 e3 29 da 51 67 91 b9 2d 9c ff	.R.K.8.) .Qg.....
0100	2d 59 62 4f 99 75 2d 4e d0 8a 8a 78 7f b1 11 5d	-Yb0-u-N ...x...]
0110	1d 77 55 db 57 89 d6 79 5d 35 aa f8 dd ee 82 08	.wU.W..y]5.....
0120	59 19 7c c8 42 05 26 ff 43 81 cb f6 fd 1c 16 39	Y. .B.&. C.....9
0130	42 cc 18 af a1 d8 90 26 ee 9c 04 81 cb dd 8e 0a	B.....&
0140	4f eb 7a 75 02 01 a3 35 a2 6e 64 50 53 f5 98 29	0.zu...5 .ndPS..)
0150	6d 6d 2f c0 06	mm/..

⬢ Payload is encrypted application data (tls.app_data), 282 bytes

?

n=131840 Len=0
=1 Win=64240 Len=0 MS
n=131840 Len=0
n=131840 Len=1360
n=131840 Len=1360
Win=64128 Len=0
Win=63744 Len=0
, Application Data
Win=63360 Len=0
Ack=1799 Win=64128 Le
721 Win=131840 Len=0
ta, Application Data
Data
Application Data
Ack=1767 Win=64128 Le
721 Win=131840 Len=0
ta, Application Data
Data
955 Win=64128 Len=0
105 Win=131840 Len=0

Reflections...

- The Internet was designed for a minimum of **reliability** and with very little thought to **security**
 - Has become the main spread vector for **malware**
- Network security
 - How to protect communications from being **intercepted** or **altered**
 - Viruses, worms, Trojan horses, spyware, spam...
 - Denial of Service Attack (DOS)

More reflections

All the protocols in this lecture send their data in clear text – this makes them easy to intercept.

Think a little about security on the Internet and how these services can be provided more securely.



Høyskolen
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